

std::zstring_view

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§ 1 Abstract

We propose a standard string view type that guarantees null-termination.

§ 2 Introduction

C++17 introduced `std::string_view`, a non-owning view of a continuous sequence of characters. It is cheap to use, offers fast operations, and replaced most uses of `const std::string&` or `const char*` as function parameters. The utility and interface of string views as well as the benefits of not having to do unnecessary `strlen` calculations on C-style strings makes string view types highly desirable for working with strings in C++.

Unlike `const std::string&` and many but not all `const char*`, `std::string_view` is not null-terminated. This allows it to have fast and cheap substring operations. However, this also means `std::string_view` is not a suitable replacement for either of the two aforementioned types whenever a null-terminated C-style string is needed. While most C++ code mostly interfaces with C++ code, it is not uncommon to need to use operating system calls, C interfaces, third-party library APIs, or even C++ standard library APIs which require null-terminated strings. Because of a lack of a desirable option for passing non-owned null-terminated strings, `std::string_view` parameters are none the less sometimes used today in cases where null-terminated strings are needed, calling `std::string_view::data` to get a `const char*`. This is, needless to say, very bug-prone.

For this reason, many C++ developers use custom `zstring_view` or `cstring_view` types which are guaranteed to be null-terminated. We propose standardizing this utility.

§ 3 Previous Papers

The idea of `zstring_view` was present even in the original paper for `string_view` (Sept 2012 - Feb 2014) <https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2014/n3921.html#null-termination>:

Another option would be to define a separate `zstring_view` class to represent null-terminated strings and let it decay to `string_view` when necessary. That's plausible but not part of this proposal.

An attempt was made in Feb 2019 to concretely propose the type (renamed to `cstring_view`) with <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2019/p1402r0.pdf>, but it failed to gain consensus in LEWG at <https://wiki.edg.com/bin/view/Wg21kona2019/P1402>. In fact, the discussion there concluded with "CONSENSUS: We will not pursue P1402R0 or this problem space." It is also worth noting that contracts were briefly contemplated in the minutes as a way of addressing the problem, however, contracts can't check for a null character safely as it wouldn't be part of the view. Similarly, other runtime checks for `string_view` null-termination are off the table.

However, in 2024 <https://wg21.link/p3081> was published which contains an aside regarding a null-terminated `zstring_view` in section 8, noting:

This is one of the commonly-requested features from the <https://github.com/microsoft/GSL> library that does not yet have a `std::` equivalent. It was specifically requested by several reviewers of this work.

During the discussion, it was made clear that `zstring_view` deserves attention on its own and it should not be happenstance introduced as part of an entirely different topic. While P3081 did not end up passing for reasons other than `zstring_view`, it is another example of the utility being seen as desirable.

We also have <https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2025/p2996r9.html>, which in its specification of functions returning a `string_view` or `u8string_view` tries its best to say it's actually supposed to be a `zstring_view`, except without naming the type for existential reasons:

Any function in namespace `std::meta` that whose return type is `string_view` or `u8string_view` returns an object `V` such that `V.data()[V.size()] == '\0'`.

As such, we do believe that there is both space for such a type, and a desire from multiple angles to have it defined in the standard library.

§ 4 Reasons to have the type

Many functions right now whether C++ standard library calls, operating system calls, or third-party library calls require null-terminated C-style strings. Absent an efficient type that can represent a non-owning view of a null-terminated string, these functions must take either a `const char*` and incur potential `strlen` overhead, take a `const std::string&` and possibly superfluously copy and allocate, take a `string_view` and make a copy, or haphazardly take a `std::string_view` with a fragile unenforceable contract that it is a view of a null-terminated string. Such a contract would be unenforceable because `pre(sv[sv.size()] == 0)` is potentially undefined behavior.

String views tend to start their life coming from a string literal or from a type that owns its internal buffer, often `std::string`. Both of these are places that can always create a `zstring_view` instead, as they both know their data is inherently null terminated.

Strings are very often just passed along with a non-owning string type all the way to where they are used as data. Some of these potential uses, such as calling `std::ofstream::write`, do not rely on the null terminator, but others, like `std::ofstream::ofstream`, do. Using `string_view` as the intermediate type loses the knowledge that a null terminator is already guaranteed to be present, requiring the developer to either make assumptions or make a copy. Another common use might be creating a stringstream from a string <https://stackoverflow.com/questions/58524805/is-there-a-way-to-create-a-stringstream-from-a-string-view-without-copying-data>.

We do not have to look far for examples of `std::string_view` being used in bug-prone ways with APIs expecting null-terminators. Searching `/\.data\(\)/ string_view language:c++ -is:fork` on GitHub code search turns up two examples on the first page:

From <https://github.com/surge-synthesizer/shortcircuit/blob/ba6ea3a2e703e4fa0ed069427aa42b668699b624/libs/md5sum/demo.cc#L37>

```
std::optional<std::string> hash_file(const std::string_view &file_name) {
    auto fd = open(file_name.data(), O_RDONLY);
    ...
}
```

From https://github.com/VisualGMQ/gmq_header/blob/cbc2853f391acc51ddd25a50d567cac404776413/log.hpp#L66

```
void log(Level level, std::string_view funcName, std::string_view filename, unsigned int line, Args&&... arg)
{
    if (level <= level_) {
        printf("[%s] [%s] [%s] [%u]", Level2Str(level).data(), filename.data(), funcName.data(), line);
        ...
    }
}
```

These two examples could be argued to be bad code or misuses of `std::string_view`, however, the fact that they are written reflects the desire for the ability to use string view types in such cases.

These problems are visible enough that we have targeted patches for specific instances of this problem in the standard - <https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2022/p2495r0.pdf> attempts to directly patch this specific example. The solution bypasses that the problem is that we're missing an unbroken type-safe chain of knowledge that the value being passed is or is not null-terminated.

The type effectively fills out the design space that exists around strings within C++. We started off with `std::string` as an owning string type, ambiguously being specified as having a null terminator or not, clarified in C++11 to definitely "have" a null terminator. C++14 added `std::string_view` to that set, offering a non-null-terminated non-owning string type. The type itself raises the question, should we add a non-null-terminated owning string type, and/or a null-terminated non-owning string type? We're proposing the last of these, leaving only the non-null-terminated owning string type as not present. We believe there is no situation to be written where a non-null-terminated owning string type would have a measurable benefit over the null-terminated owning string type that we have, and as such it is not worth the added complexity.

Without them we retain the question we had before - `const char*`, `const std::string&` or `string_view`? The first loses lots of type safety and potentially requires redundant `strlen` computation elsewhere in the software, the second requires it to be a `std::string` but retains null-termination knowledge, the last offers the ability to send any string type constructible to a `string_view`, but loses any knowledge of null termination. For the common case of passing along "some string input" to a function that ends up requiring null termination, the proposed `zstring_view` is the only type that properly captures it.

Since the first draft of this paper the use of `cstring_view` and `zstring_view` on Github has grown from 1.9k to 2.1k, indicating active use, and in many cases people adding new implementations of the type. Many contain the subtle bugs that we highlight in this paper, illustrating why it is a good idea to add the type to the standard.

§ 5 Reasons not to have the type

In an ideal world, we could actually fix the operating systems and third-party libraries to accept pointer-and-size strings in all places, removing the need for null termination to exist, and for null termination propagation to be relevant. Sometimes, in particular from environments where this may not be unrealistic in a subset, the argument is voiced that `zstring_view` does not offer any benefits.

Adding the type complicates the type system around strings by having more options for string types. This is not as much of an argument as it seems, since for each site a single type is the most appropriate, and the only places that would differ are those where the difference can make a meaningful performance impact - ie, the exact kind of tight loop where the type is useful for being able to make the difference.

§ 6 Subtle details

§ 6.1 `zstring_view` and `string_view` relation

Initially we expected that `zstring_view` may be implementable in terms of `string_view`. This is not actually true; in particular it is not well-formed to use `string_view`'s `operator=` to assign a non-null-terminated `string_view` to a `zstring_view`. As such, there can not be an inheritance relation between the two.

`zstring_view` is transparently convertible to a `string_view`. For the conversion operator, there is no `operator=` that could affect the original `zstring_view`, and as such it is a correct change to do. It in particular makes it so that any use of `zstring_view` in places where `string_view` is expected or desired will work without additional work.

This brings into reality an overload ambiguity:

```
void handle(string_view v);
void handle(zstring_view v);
handle("Hello World!");
```

This is now an ambiguous function call. It is not a new problem; <https://godbolt.org/z/c31e31fKW> shows that the exact same problem already happens with regular `std::string` and `std::string_view`. The reason is somewhat fundamental; all three types model the concept "string" and *are* ambiguous. The user should be clear about which properties of string it's expecting. Adding this new type only adds to the vocabulary the option to say "non-owning null-terminated", giving the user better choices rather than complicating it.

§ 6.1.1 Deprioritizing one overload

When one overload is preferred, but others should exist, the less preferred ones can be marked with `requires true`. This lifts one overload up out of the overload set for ambiguous matches, while leaving the rest to exist.

§ 6.1.2 Tag type forced conversions for exact matches

<https://godbolt.org/z/8qssnq8rf> and <https://godbolt.org/z/xG7955hf9> illustrate the idea of using a wrapping template to make one overload stand out.

§ 6.1.3 Explicitly adding a function to disambiguate manually

<https://godbolt.org/z/9zoGEh1cv> demonstrates adding an overload that matches anything that isn't an exact match, which indirects to a function that hand-picks an overload.

§ 6.2 Construction

`std::basic_string_view` offers a handful of constructors:

```
constexpr basic_string_view(const charT* str); // (1)
```

The basic constructor from an unadorned `charT*` risks taking a non-null-terminated string and searching for the null terminator. For `zstring_view` this constructor is as safe as it is for `string_view`, since we need to have a null terminator anyway.

```
constexpr basic_string_view(const charT* str, size_type len); // (2)
```

This constructor conceptually offers a $O(1)$ construction time. For `zstring_view` we add a contract that asserts that `str[len] == '\0'`.

```
template<class It, class End>
constexpr basic_string_view(It begin, End end); // (3)
```

This constructor is problematic to support, as we have to check the end iterator to ensure the string is null-terminated, but the end iterator cannot be assumed to be dereferencable.

```
template<class R>
constexpr explicit basic_string_view(R&& r); // (4)
```

This constructor is difficult to support as well; we need to check that after the end of the range there is a null terminator, which is by definition accessing out of the range.

```
basic_string_view( std::nullptr_t ) = delete; // (5)
```

We should not support constructing a string from nullptr.

§ 6.3 Member functions on `string_view` that return a `string_view`

In some cases the functions can be replicated on `zstring_view` with the return type being a `zstring_view`, while in some cases the return type loses the ability to guarantee null termination, and should still return a `string_view`. The types are somewhat entangled.

§ 6.3.1 `substr`

The `substr` function is changed from a single function with a default argument, to two functions, one with one and one with two arguments. The one-argument `substr` always retains the end (identical to `string_view`'s `substr`) and returns a `zstring_view`; the two-argument `substr` at least in a conceptual sense chops off at least the null terminator from the end, and should always return a `string_view`.

§ 6.3.2 `remove_prefix`

`remove_prefix` drops characters from the front of the string it's called on. Behavior fits.

§ 6.3.3 `remove_suffix`

`remove_suffix` changes the string in-situ, and as such is unimplementable on `zstring_view`. It should be a function marked as `=delete` with a reason pointing users to use the two-argument `substr` function instead, which returns a `string_view`.

§ 7 Bikeshedding

Null-terminated string view types are typically named `zstring_view` (N3921, P3081, GSL) or `cstring_view` (P1402). `cstring_view` follows the `std::string::c_str()` nomenclature while `zstring_view` has some establishment and recognizability.

Github code search shows similar popularity between https://github.com/search?q=%2F%5Cbzstring_view%5Cb%2F%20language%3Ac%2B%2B%20-is%3Afork&type=code (1.2k results as of the time of writing) and https://github.com/search?q=%2F%5Cbzstring_view%5Cb%2F+language%3Ac%2B%2B-is%3Afork&type=code (680 results as of the time of writing). In refreshing this paper, it appears that `cstring_view` has gained ~200 added results, while `zstring_view` only added 8, hinting at a popular preference for the former name.

§ 8 Reference Implementation

A reference implementation is at https://github.com/jeremy-rifkin/zstring_view.

§ 9 NVIDIA Experience (moved from p3710)

NVIDIA implemented `zstring_view` independently, with almost identical features. This is described in <https://wg21.link/p3710>, now merged into this paper. We also implemented some additional features described in <https://wg21.link/p3566>.

Ideally we wanted our input parameters to all be `string_view` and all the output parameter to be `zstring_view` (which gives more flexibility).

In practice, there are exceptions on both sides:

- When a null-terminated parameter is needed for internal reasons outside of our control, i.e. when interacting with system calls or third-party libraries, it's preferable to have a `zstring_view` parameter.
- When returning a part of an internal string, e.g. when extracting the file name from a full path, it's more convenient to just return a `string_view`.

Nonetheless, having a large codebase, we knew we couldn't just rewrite the entire codebase to apply this change, so we used `zstring_view` as a tool for steering our codebase in the ideal direction described above.

Initially, we used as `zstring_view` as a drop-in replacement for `const char*` parameters and return values of our APIs, without worrying too much whether or not the function expects a null terminator. As a sidenote, for this migration we asked our developers to be intentional in using `.data()` or `.c_str()`, and use `.data()` when a null-terminator is not expected, and only use `.c_str()` when

the string is required to be terminated. If respected, this rule allows us to just try changing a parameter from `zstring_view` to `string_view` and verify whether or not a function is relying on some parameter to be null-terminated. This operation can be done incrementally, one function at a time.

Once this was done, we started from the "deepest" functions to analyze their usage, and replace their parameters `zstring_view` to `string_view`, or rewrite them to allow for `string_view` parameters. We proceeded upwards in the call chain, and did the same. Again, this can be done in multiple passes.

What we observed is that, in some case, this might not only make the code safer, but also enable new optimizations.

Here follows an example of a parameter upgrade how we implemented these changes, step-by-step, with an example of how this can lead to more performant and safer code at the end of the process.

```
void f(const char* x) {
    external_library::g1(x);
}

void f1() {
    std::string a1 = "test";
    f(a1.c_str());
}

void f2() {
    constexpr char a2[] = "test";
    f(a2);
}

void f3() {
    const char* a3 = "test";
    f(a3);
}

void f4() {
    string_view a4 = "abcde";
    string_view subtext = a4.substr(2, 2); // no null terminator at the end of `subtext`

    f(string{subtext}); // create a temporary to add the terminator
}
```

In this example, `external_library::g1` is a placeholder for some generic computation that happens on the string.

Initially, we update the API we will change `f` to:

```
void f(zstring_view x) {
    external_library::g1(x.c_str());
}
```

All the functions now have well-defined memory ranges (no unbounded string).

We can now then look for alternatives, and find out there's a different API for the `external_library::g1` we can use, e.g. `external_library::g2(char*, size_t)`, that doesn't require a null-terminated sequence.

```
void f(string_view x) {
    external_library::g2(x.data(), x.size());
}
```

As next step, we can check all the usages, and see if we can get rid of some extra code, for example:

```
void f4() {
    string_view a4 = "abcde";
    string_view subtext = a4.substr(2, 2); // no null terminator at the end of `subtext`

    f(subtext); // f now accepts a string_view, no need for temporary
}
```

This results in a better-optimized code at the end (less need for temporary strings).

NOTE: The sequence of changes in our codebase was different, because our codebase includes the changes proposed in <https://wg21.link/p3566>, so some of the intermediate steps relying on implicit `char*` -> `string_view` and `char*` -> `zstring_view` conversions require additional changes, and intermediate steps marking conversions with the proposed `unsafe_length` tag.

§ 10 Standard Library Changes

There are a handful of interfaces in the standard library which take `const string&` and really want a `zstring_view` or possibly a `string_view`. These include:

- Constructors for `std::exception` types, `system_error`, `format_error`, `ios_base::failure`, and `std::filesystem::filesystem_error`
- The `stoi` family of functions
- `random_device::random_device`
- Lots of locale interfaces
- `basic_filebuf::open`, `basic_ifstream::basic_ifstream`, `basic_ifstream::open`, `basic_ofstream::basic_ofstream`, `basic_ofstream::open`, `basic_fstream::basic_fstream`, and `basic_fstream::open`

There are further interfaces which take `const char*` and could have overloads taking a `zstring_view` or `string_view`.

We do not propose any such changes in this paper but would like to investigate this in a follow-up paper.

§ 11 Polls

The first question to LEWG is, do we want to have the `zstring_view` type in the standard library? So far it looks like voting against the type leads to people implementing their own, including C++ committee members "implementing" their own `zstring_view` in wording. We also have a few direction questions with regards to either "fixing" string interface, or keeping the interfaces synchronized.

§ 11.1 We encourage further work on the `zstring_view` proposal for the C++29 timeframe.

If so, we will return at the next meeting (Kona 2025) with the wording worked out and checked with a Core representative, with the intent of merging `zstring_view` early in the C++29 cycle. If not, we will retire this paper.

§ 11.2 We prefer the name `cstring_view`

The bikeshedding mentioned above. At this point we can still choose which name to use. The authors have no preference.

- `cstring_view` is more common on Github at the time of writing, at a factor of about 2:1
- `cstring_view` matches existing `std::string` interface nomenclature (`c_str()`).
- `zstring_view` is the most recently proposed name in <https://wg21.link/p3081>

§ 11.3 We prefer `zstring_view` to remove `char_traits` from its interface.

SG16 has long held that `char_traits` offer no value and are sometimes abused. If possible they would propose to remove it altogether [P3681](#). This paper however is trying to add `zstring_view` to the standard without changing the direction for `char_traits`, and as such it is not proposing to remove `char_traits`.

Removing it anyway would necessarily lose information on implicit conversion from `std::string` through `std::zstring_view` to `std::string_view`, as the `char_traits` that were present on the string are no longer possible to transfer to the `string_view`. There was a discussion in SG16 with regards to `char_traits` needing to be removed, needing to be kept, or whether no objection existed to retaining it for compatibility, and the following poll was taken:

Poll 1: P3655R0: No objection to use of `std::char_traits` for consistency and compatibility with `std::string_view`.

SF	F	N	A	SA
7	1	0	0	0

Strong consensus. Attendees: 8 (no abstentions)

The paper authors hold no strong opinion on whether `char_traits` should be kept or removed. Pragmatically we propose to keep `char_traits` to make the conversion through `zstring_view` lossless, and to treat removal of `char_traits` support as a uniform thing to be applied to all string types. [P3681](#) disagrees, however, and we hope to see it up for discussion to reduce user-visible churn.

§ 11.4 We prefer the `zstring_view` interface to add an array constructor

```
template <size_t N>
constexpr basic_zstring_view(const charT (&str)[N]);
```

The advantage of adding this function is that it is a clean constructor to convert from a string literal with compile-time known size. The downsides are that it will create an interface difference between `string_view` and `zstring_view`.

§ 11.5 We prefer to disallow contained NULs

String and `string_view` both support having a string with embedded NUL characters. On one hand it makes complete sense to do the same for `zstring_view`. On the other hand, the value of `zstring_view` lies in knowing it has a null terminator at `size()`, and while allowing embedded NULs would not strictly violate that - there still is a NUL terminator there - it would logically break it, as there would be a NUL before that that effectively terminates the string.

This gives rise to the possibility for bugs where strings are being used by their `strlen(x.c_str())` length, and elsewhere by their `x.size()`. These things are likely enough to have <https://cwe.mitre.org/data/definitions/135.html> and <https://cwe.mitre.org/data/definitions/179.html>, quote:

An attacker could include dangerous input that bypasses validation protection mechanisms which can be used to launch various attacks including injection attacks, execute arbitrary code or cause other unintended behavior.

Consider the following call:

```
std::zstring_view user_command = "ls\\0rm -rf /";
validate_command(user_command.c_str()); // older validation function that does its own strlen()
evaluate_command(user_command); // newer function that uses user_command.size()
```

We can prevent these bugs at this stage by disallowing contained NULs in the NUL-terminated string view type. But it is a breaking change to existing expectation.

A strong argument against is that constructing a `zstring_view` from this string literal will behave differently than constructing a `string_view` would, and the invariant that `zstring_view` is null-terminated is not broken with an embedded NUL.

§ 12 Wording

This wording section is currently mostly high-level and synopsis changes. We defer full wording until feedback from LEWG.

Borrowing extensively from existing `string_view` wording.

Add to `[format.formatter.spec]` 2.2:

```
template<class traits>
struct formatter<basic_zstring_view<charT, traits>, charT>;
```

Add to `[format.formatter.spec]` 4.1:

```
template<class traits>
struct formatter<basic_zstring_view<char, traits>, wchar_t>;
```

Update [string.view.general](#) note 1 to mention conversion from `basic_zstring_view` to `basic_string_view`.

Add to [basic.string.general](#):

```
constexpr operator basic_zstring_view<charT, traits>() const noexcept;
```

Wording on additions deferred until a later revision.

§ 12.1 Zstring View Classes [zstring.view]

§ 12.1.1 General [zstring.view.general]

The class template `basic_zstring_view` describes an object that can refer to a constant contiguous sequence of char-like ([strings.general](#)) objects with the first element of the sequence at position zero. In the rest of [string.view](#), the type of the char-like objects held in a `basic_string_view` object is designated by `charT`.

In all cases, `[data(), data() + size())` is a valid range and `data() + size()` points at an object with value `charT()` (a "null terminator").

[Note 1: `basic_zstring_view<charT, ...>` is primarily useful when working with C APIs which expect null terminated strings. - end note]

§ 12.1.1.1 Header `` synopsis [zstring.view.synop]

```
// mostly freestanding
#include <compare> // see [compare.syn]

namespace std {
    // [zstring.view.template], class template basic_zstring_view
```



```

template<class charT, class traits = char_traits<charT>>
class basic_zstring_view; // partially freestanding

template<class charT, class traits>
constexpr bool ranges::enable_view<basic_zstring_view<charT, traits>> = true;
template<class charT, class traits>
constexpr bool ranges::enable_borrowed_range<basic_zstring_view<charT, traits>> = true;

// [zstring.view.comparison], non-member comparison functions
template<class charT, class traits>
constexpr bool operator==(basic_zstring_view<charT, traits> x,
                           type_identity_t<basic_zstring_view<charT, traits>> y) noexcept;
template<class charT, class traits>
constexpr /* see below */
operator<=(basic_zstring_view<charT, traits> x,
            type_identity_t<basic_zstring_view<charT, traits>> y) noexcept;

// [zstring.view.io], inserters and extractors
template<class charT, class traits>
basic_ostream<charT, traits>&
operator<<(basic_ostream<charT, traits>& os,
            basic_zstring_view<charT, traits> str); // hosted

// basic_zstring_view typedef-names
using zstring_view = basic_zstring_view<char>;
using u8zstring_view = basic_zstring_view<char8_t>;
using u16zstring_view = basic_zstring_view<char16_t>;
using u32zstring_view = basic_zstring_view<char32_t>;
using wzstring_view = basic_zstring_view<wchar_t>;

// [zstring.view.hash], hash support
template<class T> struct hash;
template<> struct hash<zstring_view>;
template<> struct hash<u8zstring_view>;
template<> struct hash<u16zstring_view>;
template<> struct hash<u32zstring_view>;
template<> struct hash<wzstring_view>;

inline namespace literals {
    inline namespace zstring_view_literals {
        // [zstring.view.literals], suffix for basic_zstring_view literals
        constexpr zstring_view operator""zsv(const char* str, size_t len) noexcept;
        constexpr u8zstring_view operator""zsv(const char8_t* str, size_t len) noexcept;
        constexpr u16zstring_view operator""zsv(const char16_t* str, size_t len) noexcept;
        constexpr u32zstring_view operator""zsv(const char32_t* str, size_t len) noexcept;
        constexpr wzstring_view operator""zsv(const wchar_t* str, size_t len) noexcept;
    }
}

```

§ 12.1.2 Class template basic_zstring_view [zstring.view.template]

§ 12.1.2.1 General [zstring.view.template.general]

```

namespace std {
    template<class charT, class traits = char_traits<charT>>
    class basic_zstring_view {
    public:
        // types
        using traits_type = traits;
        using value_type = charT;
        using pointer = value_type*;
        using const_pointer = const value_type*;
        using reference = value_type&;
        using const_reference = const value_type&;
        using const_iterator = implementation-defined; // see [zstring.view.iterators]
        using iterator = const_iterator;
        using const_reverse_iterator = reverse_iterator<const_iterator>;

```



```

using reverse_iterator      = const_reverse_iterator;
using size_type             = size_t;
using difference_type       = ptrdiff_t;
static constexpr size_type npos = size_type(-1);

// [zstring.view.cons], construction and assignment
constexpr basic_zstring_view() noexcept;
basic_zstring_view(const basic_zstring_view&) noexcept = default;
basic_zstring_view& operator=(const basic_zstring_view&) noexcept = default;
constexpr basic_zstring_view(const charT* str) noexcept
    pre(str != nullptr);
constexpr basic_zstring_view(const charT* str, size_type len) noexcept
    pre (str != nullptr)
    pre (str[len] == '\0');
template <size_t N>
constexpr basic_zstring_view(const charT (&str)[N]) noexcept
    pre (traits::find(str, N, '\0') != nullptr);
basic_zstring_view(nullptr_t) = delete;

// [zstring.view.iterators], iterator support
constexpr const_iterator begin() const noexcept;
constexpr const_iterator end() const noexcept;
constexpr const_iterator cbegin() const noexcept;
constexpr const_iterator cend() const noexcept;
constexpr const_reverse_iterator rbegin() const noexcept;
constexpr const_reverse_iterator rend() const noexcept;
constexpr const_reverse_iterator crbegin() const noexcept;
constexpr const_reverse_iterator crend() const noexcept;

// [zstring.view.capacity], capacity
constexpr size_type size() const noexcept;
constexpr size_type length() const noexcept;
constexpr size_type max_size() const noexcept;
[[nodiscard]] constexpr bool empty() const noexcept;

// [zstring.view.access], element access
constexpr const_reference operator[](size_type pos) const
    pre (pos <= size());
constexpr const_reference at(size_type pos) const;
constexpr const_reference front() const;
constexpr const_reference back() const;
constexpr const_pointer data() const noexcept;
constexpr const_pointer c_str() const noexcept;

operator basic_string_view<charT, traits>() const noexcept;

// [zstring.view.modifiers], modifiers
constexpr void remove_prefix(size_type n);
constexpr void remove_suffix(size_type n) = delete("cannot remove_suffix in-place on zstring_view while");
constexpr void swap(basic_zstring_view& s) noexcept;

// [zstring.view.ops], zstring operations
constexpr size_type copy(charT* s, size_type n, size_type pos = 0) const;

constexpr basic_string_view<charT, traits> substr(size_type pos = 0, size_type n = npos) const;

constexpr int compare(basic_string_view<charT, traits> s) const noexcept;
constexpr int compare(size_type pos1, size_type n1, basic_zstring_view s) const;
constexpr int compare(size_type pos1, size_type n1, basic_zstring_view s,
    size_type pos2, size_type n2) const;
constexpr int compare(const charT* s) const
    pre (s != nullptr);
constexpr int compare(size_type pos1, size_type n1, const charT* s) const
    pre (s != nullptr);
constexpr int compare(size_type pos1, size_type n1, const charT* s, size_type n2) const
    pre (s != nullptr);

constexpr bool starts_with(basic_string_view<charT, traits> x) const noexcept;

```

```

constexpr bool starts_with(charT x) const noexcept;
constexpr bool starts_with(const charT* x) const
    pre (x != nullptr);
constexpr bool ends_with(basic_string_view<charT, traits> x) const noexcept;
constexpr bool ends_with(charT x) const noexcept;
constexpr bool ends_with(const charT* x) const
    pre (x != nullptr);

constexpr bool contains(basic_string_view<charT, traits> x) const noexcept;
constexpr bool contains(charT x) const noexcept;
constexpr bool contains(const charT* x) const
    pre (x != nullptr);

// [zstring.view.find], searching
constexpr size_type find(basic_string_view<charT, traits> s, size_type pos = 0) const noexcept;
constexpr size_type find(charT c, size_type pos = 0) const noexcept;
constexpr size_type find(const charT* s, size_type pos, size_type n) const
    pre (s != nullptr);
constexpr size_type find(const charT* s, size_type pos = 0) const
    pre (s != nullptr);
constexpr size_type rfind(basic_string_view<charT, traits> s, size_type pos = npos) const noexcept;
constexpr size_type rfind(charT c, size_type pos = npos) const noexcept;
constexpr size_type rfind(const charT* s, size_type pos, size_type n) const
    pre (s != nullptr);
constexpr size_type rfind(const charT* s, size_type pos = npos) const
    pre (s != nullptr);

constexpr size_type find_first_of(basic_string_view<charT, traits> s, size_type pos = 0) const noexcept;
constexpr size_type find_first_of(charT c, size_type pos = 0) const noexcept;
constexpr size_type find_first_of(const charT* s, size_type pos, size_type n) const
    pre (s != nullptr);
constexpr size_type find_first_of(const charT* s, size_type pos = 0) const
    pre (s != nullptr);
constexpr size_type find_last_of(basic_string_view<charT, traits> s, size_type pos = npos) const noexcept;
constexpr size_type find_last_of(charT c, size_type pos = npos) const noexcept;
constexpr size_type find_last_of(const charT* s, size_type pos, size_type n) const
    pre (s != nullptr);
constexpr size_type find_last_of(const charT* s, size_type pos = npos) const
    pre (s != nullptr);
constexpr size_type find_first_not_of(basic_string_view<charT, traits> s, size_type pos = 0) const noexcept;
constexpr size_type find_first_not_of(charT c, size_type pos = 0) const noexcept;
constexpr size_type find_first_not_of(const charT* s, size_type pos,
                                     size_type n) const
    pre (s != nullptr);
constexpr size_type find_first_not_of(const charT* s, size_type pos = 0) const
    pre (s != nullptr);
constexpr size_type find_last_not_of(basic_string_view<charT, traits> s,
                                     size_type pos = npos) const noexcept;
constexpr size_type find_last_not_of(charT c, size_type pos = npos) const noexcept;
constexpr size_type find_last_not_of(const charT* s, size_type pos,
                                     size_type n) const
    pre (s != nullptr);
constexpr size_type find_last_not_of(const charT* s, size_type pos = npos) const
    pre (s != nullptr);

private:
    const_pointer data_;           // exposition only
    size_type size_;              // exposition only
};
}

```

Wording on members deferred until a later revision.